Iago Caldentey Palmer

Software Developer – C++ · Real-time Simulation · Aerospace & Defence

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Spanish (EU citizen) · Resident in Switzerland · Available immediately



PROFILE

Software developer with over two years of professional C++ development on Linux for real-time systems, most recently on an Airbus Defence and Space programme. On SIRTAP, contributed to the architecture of the custom rendering engine for the safety-critical cockpit displays, designed a large, highly flexible occlusion system and reworked GPU submission from thousands of draw calls per frame to a single one, with every change developed, documented and verified to aerospace software standards. Previously developed low-level systems, gameplay systems and a suite of internal development tools for a proprietary C++ game engine. Builds real-time simulations: an F-16 simulator in VR, physics simulators and a one-million-agent simulation that led to a CEIG 2025 paper. Works in C++, Python and Ada. Based in Switzerland, EU citizen, available immediately and ready to relocate.

PROFESSIONAL EXPERIENCE

Rendering and Software Engineer – Cockpit Display Systems (Airbus SIRTAP) 01.2026 – 09.2026

Orbital Critical Systems, external contractor for Airbus Defence and Space · Getafe (Madrid), Spain

- Contributed to the architecture of a custom rendering engine tailored to airborne cockpit systems.
- Designed and implemented a large, highly flexible occlusion system handling the many overlapping display elements and layouts of the cockpit.
- Developed, documented and verified code in line with aerospace software development standards throughout the lifecycle; work consistently review-ready and traceable.
- Reworked how cockpit displays are submitted to the GPU, reducing rendering cost from thousands of draw calls per frame to a single one, giving the display system substantial frame-time headroom.
- Designed and implemented real-time rendering pipelines for safety-critical HMI displays, meeting strict visual-accuracy and performance requirements under airborne certification constraints.
- Identified and optimised costly CPU-side processing in the rendering engine using the Valgrind toolset (Callgrind, Cachegrind, Memcheck).
- Worked with SCADE Display and the ARINC 661 standard, bridging display definition and low-level rendering.
- Captured and analysed GPU calls with Apitrace to diagnose rendering and performance issues.

C++, OpenGL, Linux, Apitrace, Valgrind, SCADE Display, ARINC 661, Git, Bash

Game, Engine and Tools Programmer 07.2024 – 12.2025

Merkur Dosniha · Madrid, Spain

- Developed low-level systems of the studio's proprietary game engine in C++ and OpenGL.
- Built a broad suite of internal engine tools: custom performance profiler, memory allocation utilities, resource manager, sound manager and audio mixer, and low-level debugging utilities.
- Developed tooling that improved team iteration speed, runtime diagnostics and engine reliability.
- Designed and implemented gameplay systems in C++ for shipped game projects, contributing to their architecture and feature set.
- Analysed disassembled code to diagnose and fix bugs and random crashes, improving engine stability.
- Collaborated with engineers, artists and designers across Spain and Germany in a multidisciplinary, cross-cultural team.

C++, OpenGL, Vulkan, RenderDoc, Python, Linux, Git, Bash

EDUCATION

Master's Degree in Computer Graphics, Video Game Development and Virtual Reality 2025 – 2026
Universidad Rey Juan Carlos, Madrid

- Real-time applications and simulations in C++ and C# with advanced shader programming and GPU optimisation; GPU computing with CUDA.
- Virtual reality development: VR rendering and interaction.
- Implemented rendering techniques and pipelines from scratch in OpenGL and Vulkan.

Bachelor's Degree in Computer Engineering, specialisation in Artificial Intelligence and Computation 2020 – 2024
Universitat de les Illes Balears (UIB), Palma

- Final project: real-time 3D bee-colony simulation game in a procedurally generated world, simulating up to one million entities using data-oriented design (Unity DOTS / ECS); led to a peer-reviewed publication at CEIG 2025.
- Algorithms, data structures, computer architecture, software engineering and machine learning.

TECHNICAL SKILLS

Languages	C++, Python, Ada, C, C#, Bash, Java, MATLAB, 68k Assembly
Other	Linux, Windows, Git, TCP/UDP networking, MQTT, RTOS, OpenCV, Arduino
Graphics	OpenGL, Vulkan, GLSL, HLSL, CUDA, Unity, SCADE Display, ARINC 661
Profiling & debugging	RenderDoc, Apitrace, Valgrind (Callgrind, Cachegrind, Memcheck), IDA

SELECTED PROJECTS

- F-16 fighter-jet simulator in VR (Unity, C#)
- Real-time physics simulators: ocean waves, cloth/springs, rigid bodies, 2D fluids (C++, OpenGL)
- Networked FPS on Arduino clients (RTOS, TCP/IP, MQTT) with own OpenGL/OpenTK engine and level editor
- Real-time rendering engine in C++ / OpenGL
- Real-time renderer in C++ / Vulkan

AWARDS & PUBLICATIONS

- Paper at CEIG 2025 (Spanish Computer Graphics Congress): "Multidisciplinary Development of an Educational 3D Simulation Game for Bee Biology using Advanced Graphics Techniques".
- 1st place, Hunger4Innovation Hackathon 2023 (Madrid), challenge set by Alstom.
- Winner of Mallorca Game Jam 2024 – Best Game, Best Art and Most Polished.

LANGUAGES

Spanish: Native · **Catalan**: Native · **English**: C1 – Cambridge C1 Advanced (2025) · **German**: Beginner – currently taking classes

REFERENCES

Agustín Vaquero Gómez – Software Team Leader, Airbus Defence and Space
Prof. Francisco José Perales López – Director UGIVIA Research Group, UIB